

Econ 2203 | International Trade and Policy in Agriculture

Department of Development Economics

Recap: The Gap in Smith's Theory

Last lecture: Adam Smith argued that countries should export goods they produce *absolutely* more efficiently.

But this raises an obvious question:

What if one country is better at producing everything?

- Should the more efficient country trade at all?
- Should the less efficient country simply import everything?
- Would the inefficient country have *any* exports?

Smith's absolute advantage gives **no answer**.

Stylised fact: The United States is more productive than Bangladesh in *both* textiles *and* aircraft. Yet the US imports textiles from Bangladesh. **Why?** → Enter David Ricardo (1817).

Ricardo's Insight: What Matters is *Relative* Cost

Ricardo's key insight was deceptively simple:

*A country should specialise in the good for which it has the **lower opportunity cost** — regardless of absolute productivity.*

Opportunity cost = what you must give up to produce one more unit of a good.

Even a country that is absolutely less efficient in *everything* can still gain from trade by exporting the good in which it is **relatively** less inefficient.

Analogy: A lawyer who types faster than her secretary still benefits from hiring the secretary — because the lawyer's time is more valuably spent on legal work. The same logic applies to nations.

The Labor-Value Model: Notation

We work with the simplest possible model:

- **Two countries:** India (I) and the World (W)
- **Two goods:** Rice (R) and Wheat (Wh)
- **One factor of production:** Labour (L)
- **Technology:** constant labour requirements per unit of output

Labour coefficients (units of labour per unit of output):

	Rice	Wheat
India	a_{LR}^I	a_{LWh}^I
World	a_{LR}^W	a_{LWh}^W

a_{LR}^I = labour hours required to produce **one unit of Rice in India**.

Numerical Example

Assume: each country has $L = 100$ workers. One worker per day produces:

Country	Rice	Wheat
India	2 units	1 unit
World	3 units	3 units

Labour coefficients: $a_{LR}^I = \frac{1}{2}$, $a_{LWh}^I = 1$; $a_{LR}^W = \frac{1}{3}$, $a_{LWh}^W = \frac{1}{3}$. The **World** has absolute advantage in both goods.

Opportunity costs:

$$OC_R^I = \frac{a_{LR}^I}{a_{LWh}^I} = \frac{1/2}{1} = \frac{1}{2} \text{ Wheat} \quad OC_R^W = \frac{a_{LR}^W}{a_{LWh}^W} = \frac{1/3}{1/3} = 1 \text{ Wheat}$$

Comparison: $OC_R^I = \frac{1}{2} < OC_R^W = 1 \Rightarrow$ India has comparative advantage in Rice.

The Comparative Advantage Condition

Formal statement:

India has a comparative advantage in Rice if and only if:

$$\frac{a_{LR}^I}{a_{LWh}^I} < \frac{a_{LR}^W}{a_{LWh}^W}$$

Equivalently, rearranging:

$$\frac{a_{LR}^I}{a_{LR}^W} < \frac{a_{LWh}^I}{a_{LWh}^W}$$

India is *relatively* more efficient in Rice than in Wheat.

Key takeaway

Absolute productivity levels (a_{LR}^I vs a_{LR}^W) do **not** determine trade patterns. **Relative** productivity — the ratio of ratios — does.

Production Possibility Frontiers

With $L = 100$ workers, each country's PPF is a straight line (constant opportunity cost):

India:

$$R + 2 \cdot Wh = 200$$

Slope = -2 (give up 2 Rice per Wheat)

World:

$$R + 1 \cdot Wh = 300$$

Slope = -1 (give up 1 Rice per Wheat)

The **slope** of the PPF = opportunity cost of Wheat.

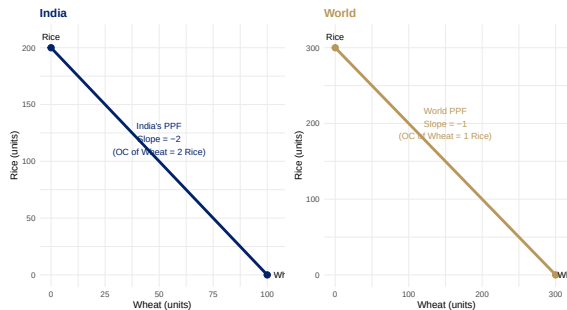


Figure 1: Production Possibility Frontiers: India and World Source: Author's illustration.

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The Terms of Trade Range

For trade to be mutually beneficial, the world price ratio must lie **between** the two autarky opportunity costs.

Autarky opportunity costs of Wheat (in terms of Rice):

$$OC_{Wh}^I = 2 \text{ Rice} \quad OC_{Wh}^W = 1 \text{ Rice}$$

The **terms of trade** (price of Wheat relative to Rice) must satisfy:

$$1 < \frac{P_{Wh}}{P_R} < 2$$

- If $P_{Wh}/P_R \leq 1$: India would not import Wheat (cheaper to produce at home)
- If $P_{Wh}/P_R \geq 2$: World would not export Wheat (too expensive)
- At any intermediate price, **both countries benefit**.

Equilibrium ToT is determined by relative demand – say $P_{Wh}/P_R = 1.5$.

Gains from Trade: India's Perspective

Mechanism:

1. India **fully specialises** in Rice → produces (0 Wheat, 200 Rice)
2. Trades Rice for Wheat at world price $P_{Wh}/P_R = 1.5$
3. **Consumption** moves to a point **above** the PPF the PPF

The trade possibilities line starts at full specialisation (0, 200) with slope = -1.5 (less steep in absolute value than the PPF slope -2).

Consumption bundle **above the PPF** = welfare gain!

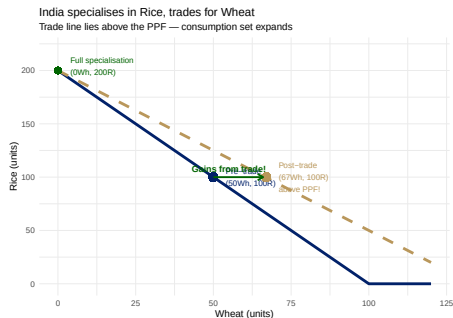


Figure 2: India: Gains from Trade — Consuming Beyond the PPF Source: Author's illustration.

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Quantifying the Gains

Before trade (autarky): India produces and consumes on the PPF at $(Wh, R) = (50, 100)$

After trade (at $P_{Wh}/P_R = 1.5$): India fully specialises $\rightarrow$ 200 Rice, trades ≈ 100 Rice for ≈ 67 Wheat:

$$(Wh, R) \approx (67, 100)$$

Welfare gain: 17 more units of Wheat at the same Rice – the **real wage** of Indian workers rises.

The algebra: budget constraint under trade: $R + 1.5 \cdot Wh = 200$

Compare with autarky constraint: $R + 2 \cdot Wh = 200$

The trade budget line is **flatter** $\rightarrow$ larger feasible consumption set.

Consumer Preferences and Equilibrium

To find the *exact* equilibrium consumption bundle, we need **indifference curves (ICs)**.

Properties of ICs:

- Downward sloping ($MRS > 0$)
- Convex to origin (diminishing MRS)
- Higher IC = higher utility

Autarky equilibrium (E1):

$$MRS = \frac{P_{Wh}}{P_R} = \frac{a_{LWh}^I}{a_{LR}^I} = 2$$

IC1 is tangent to the PPF at E1.

Free-trade equilibrium (E2):

IC2 is tangent to the **trade possibilities line** at E2.

Since IC2 is above IC1: **trade raises welfare**.

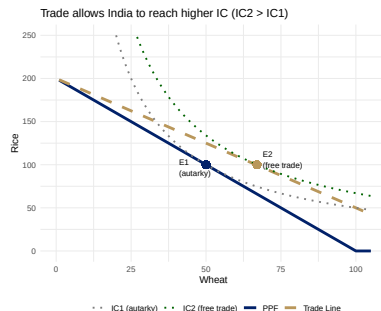


Figure 3: Free Trade Equilibrium: IC tangent to Trade Possibilities Line
Line Source: Author's illustration.

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Utility Maximisation Under Trade

Formal optimisation:

India maximises $U(R, Wh)$ subject to the trade budget constraint:

$$\max_{R, Wh} U(R, Wh) \quad \text{s.t.} \quad P_R \cdot R + P_{Wh} \cdot Wh = P_R \cdot \bar{R}$$

where $\bar{R} = 200$ (full specialisation output).

First-order condition:

$$\frac{MU_R}{MU_{Wh}} = \frac{P_R}{P_{Wh}} = \frac{1}{1.5}$$

Separation of production and consumption decisions:

Under free trade, India:

1. **Produces** where slope of PPF $= P_R/P_{Wh} \rightarrow$ full specialisation in Rice
2. **Consumes** where MRS $= P_R/P_{Wh} \rightarrow$ tangency of IC with trade line

These two points generally differ — the gap is filled by **trade flows**.

Key Assumptions of the Ricardian Model

Assumption	Implication
One factor (labour)	Ignores capital, land
Constant returns to scale	PPF is linear
Perfect factor mobility within country	Instantaneous reallocation
Immobile factors across countries	No international migration
No transport costs	TOT fully determines trade
Perfect competition	No strategic behaviour

Remarkably robust predictions:

- Countries do specialise along lines of comparative advantage
- Trade raises real incomes on average
- Even the “least competitive” country has something to export
- Cannot explain intra-industry trade (→ new trade theory, Lecture 6)
- **Extensions:** Many goods (Dornbusch-Fischer-Samuelson chain); technology changes shift comparative advantage over time

Measuring Comparative Advantage: The Balassa Index

The **Revealed Comparative Advantage (RCA)** index (Balassa, 1965):

$$RCA_{ij} = \frac{X_{ij}/X_{i.}}{X_{.j}/X_{..}}$$

where:

- X_{ij} = country i 's exports of good j
- $X_{i.}$ = country i 's total exports
- $X_{.j}$ = world exports of good j
- $X_{..}$ = world total exports

Interpretation:

- $RCA_{ij} > 1$: country i has a comparative advantage in good j
- $RCA_{ij} < 1$: country i has a comparative disadvantage in good j

India's RCA in Agricultural Products

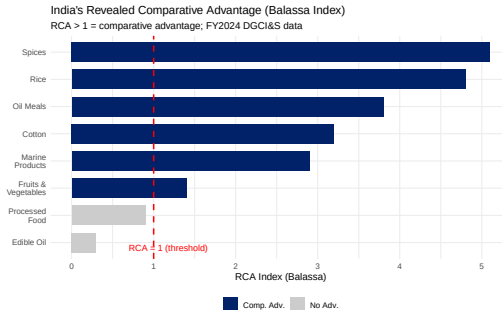


Figure 4: India: Revealed Comparative Advantage in Agricultural Products (FY2024) Source: Author's calculations using DGCI&S / UN Comtrade data.

India: Rice as the Flagship Export

India's Rice sector — Ricardo in practice:

- India is the **world's largest rice exporter** (>22% of global exports, FY2024)
- Rice cultivation is labour-intensive — India's abundant factor
- **Relative** labour productivity in rice > relative labour productivity in capital-intensive goods

Policy tension: September 2023: India imposed a **25% export duty** on non-Basmati white rice → global rice prices spiked 15%.

$$P_R^{\text{world}} \uparrow \Rightarrow \text{India's ToT deteriorate for future exporters}$$

The ban is welfare-reducing under comparative advantage theory: India surrenders gains from trade to achieve food security goals. **Central tension:** Static efficiency ↔ food security; comparative advantage ↔ strategic reserves; export competitiveness ↔ domestic price stability.

Comparative Advantage: Summary

Core results of the Ricardian Model:

1. Trade is driven by **relative**, not absolute, productivity differences
2. Comparative advantage condition: $\frac{a_{LR}^I}{a_{LWh}^I} < \frac{a_{LR}^W}{a_{LWh}^W}$
3. Gains from trade: consumption bundle moves **beyond the PPF**
4. Terms of trade must lie between autarky opportunity costs: $OC_R^W < \frac{P_R}{P_{Wh}} < OC_R^I$
5. Both countries gain — even the absolutely less productive one

Empirical verdict: Labour productivity differences strongly predict export patterns — Costinot et al. (2012) find $R^2 \approx 0.9$ across countries. India's rice exports (RCA = 4.8) confirm comparative advantage. Complete specialisation is rarely observed in practice, and short-run distributional losses are documented — but aggregate welfare gains are robust.

What Ricardo cannot explain: *Why* labour productivity differs (→ H-O model); income distribution effects (→ Stolper-Samuelson); intra-industry trade (→ Lecture 6)

Further Reading

- *International Economics* — Salvatore (Ch. 3)
- *International Economics* — Appleyard & Field (Ch. 3)
- RBI/DGCI&S/APEDA databases for latest data

Key Data Sources

- DGCI&S: India's merchandise trade
- RBI: Balance of payments data
- APEDA: Agricultural export statistics
- WTO: Tariff and trade databases